video 90.
12:00

# Diferencijalne jednadžbe

## ~ kontinuirani sustavi ~

Dif. jed: $$\frac{d^N y}{dt^N} + a_1 \frac{d^{N-1} y}{dt^{N-1}} + \dots + a_{N-1}\frac{dy}{dt} + a_N y(t) =$$

$$= b_{N-M}\frac{d^M u}{dt^M} + b_{N-M+1}\frac{d^{M-1} u}{dt^{M-1}} + \dots + b_{N-1}\frac{du}{dt} + b_N u(t)$$

→ za realne fizikalne sustave je $M \leq N$ pa jednadžbu u najopć. slučaju $M=N$ pišemo:

$$\frac{d^N y}{dt^N} + a_1 \frac{d^{N-1} y}{dt^{N-1}} + \dots + a_{N-1}\frac{dy}{dt} + a_N y(t) = b_0 \frac{d^N u}{dt^N} + b_1 \frac{d^{N-1} u}{dt^{N-1}} + \dots + b_{N-1}\frac{du}{dt} + b_N u(t)$$

⇝ uvodenjem operatora derivirana $D$ koji predstavlja operaciju deriviranja $\frac{d}{dt}$

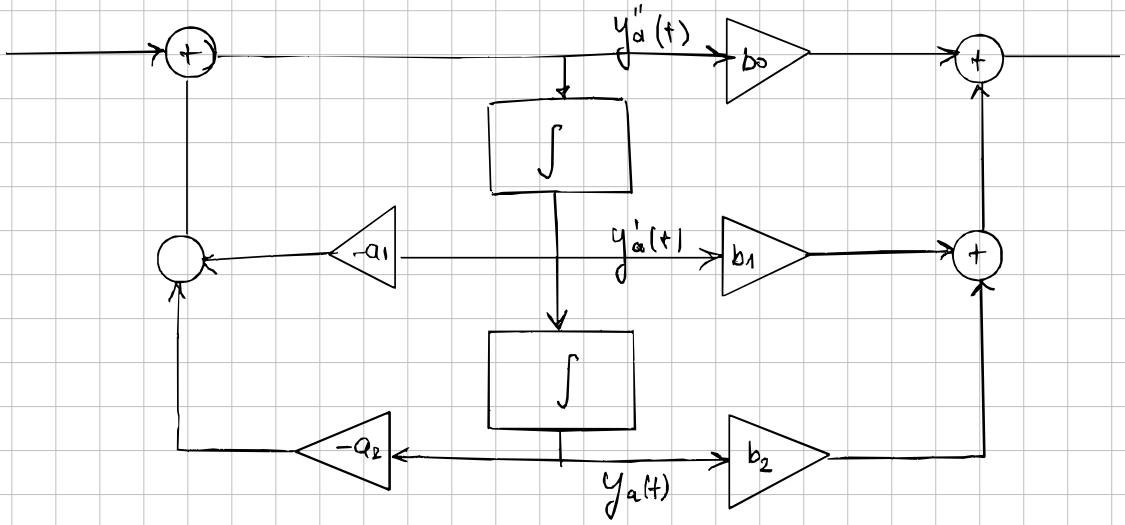
$$\underbrace{\left[D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N\right]}_{A(D)} y(t) = \underbrace{\left(b_0 D^N + b_1 D^{N-1} + \dots + b_{N-1} D + b_N\right)}_{B(D)} u(t)$$

$$\Rightarrow \forall t \in \mathbb{R} \quad A(D)\, y(t) = B(D)\, u(t)$$

**Blokovski dijagram DJ:** → pomoćna II realizacija

$$A(D)\, y_a(t) = u(t) \longrightarrow y_a''(t) + a_1 y_a'(t) + a_2 y_a(t) = u(t)$$

$$y(t) = B(D)\, y_a(t) \longrightarrow y(t) = b_0 y_a''(t) + b_1 \cdot y_a'(t) + b_2 y_a(t)$$



# Linearni vremenski stalni sustavi – model U-I

→ klasično imaju homogeno i partikularno rješenje

**Primjer:** $y''(t) + 2y'(t) + 5y(t) = u(t)$ → pored $u(t)$ je uvijek $b_N$

$N=2$

$a_0 = 1$, $a_1 = 2$, $a_3 = 5$ → $b_2 = 1$

$\forall\, b_1 = \emptyset$ $b_0 = \emptyset$

**HOM**

$y_h(t) = C \cdot e^{st}$

$y'_h(t) = C \cdot s \cdot e^{st}$

$y''_h(t) = C \cdot s^2 \cdot e^{st}$

→ * $y^N_h(t) = C s^N \cdot e^{st}$

* $u(t) = 5\mu(t)$

(budući da je pobuda konstanta)

$y_p(t) = K$ (kasnije ćemo množiti sa $\mu$)

$\Rightarrow y''(t) + 2y'(t) + 5y(t) = 0$

$C \cdot s^2 \cdot e^{st} + 2C \cdot s e^{st} + 5 \cdot C \cdot e^{st} = 0 \quad / : Ce^{st}$

$s^2 + 2s + 5 = 0$ ···> karakteristični polinom → $s_{1,2} = \dots = -1 \pm 2j$

karakt. jednadžbi

$$\boxed{y_h = C_1 e^{(-1+2j)t} + C_2 e^{(-1-2j)t}}$$

* intermezzo: da je npr. bilo zadano: $s_{1,2} = 5$ $s_3 = 6$

↳ $y_h(t) = C_1 e^{5t} + C_2 t \cdot e^{5t} + C_3 e^{6t}$

**PART** $u(t) = 5\mu(t)$

$y_p(t) = K$

$y'_p(t) = 0$

$y''_p(t) = 0$

$y''_p(t) + 2y'_p(t) + 5y_p(t) = 5$

$0 + 2 \cdot 0 + 5 \cdot K = 5 \longrightarrow K = 1$

$$\boxed{y_p(t) = 1, \quad t \geq 0}$$

| POBUDA | PARTIKULARNO |
|---|---|
| $A$ | $K$ |
| $At^M$ | $K_0 + K_1 t + K_2 t^2 + \dots + K_M t^M$ |
| $Ae^{\xi t}$ | $Ke^{\xi t}$, $\xi \neq s_i$ (različito) |
| $Ae^{\xi t}$ | $Kt^{\underline{i}} e^{\xi t}$, $\xi = s_i$ (jednako) |
| $A\cos(\omega_0 t)$ | $K_1 \cos(\omega_0 t) + K_2 \sin(\omega_0 t)$ |
| $A\sin(\omega_0 t)$ | |

(općenito)

Najgori mogući slučaj: zdesna su isto derivacije

* pokuda je kauzalna

$$y''(t) + a_1 y'(t) + a_2 y(t) = b_0 u''(t) + b_1 u'(t) + b_2 u(t)$$

$u(0^-) = 0$, $u'(0^-) = 0$, $u''(0^-) = 0$

* nemamo impuls na ulazu. $\rightarrow \delta(t) \neq u(t)$

$y(0^-) \neq 0$; $y'(0^-) \neq 0 \rightarrow$ zadani poč. uvjeti

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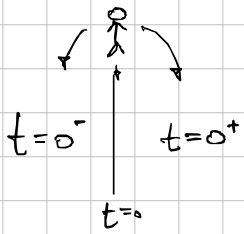
$y(0^+) = ?$ $\quad y'(0^+) = ?$




$\rightarrow$ integriramo od $(0^-)$ $\rightarrow$ malo prije djelovanja nožnog stupa pa do sadašnjosti

$$\Rightarrow \int_{0^-}^{t} y''(\tau)\,d\tau + a_1 \int_{0^-}^{t} y'(\tau)\,d\tau + a_2 \int_{0^-}^{t} y(\tau)\,d\tau =$$

$$= b_0 \int_{0^-}^{t} u''(\tau)\,d\tau + b_1 \int_{0^-}^{t} u'(\tau)\,d\tau + b_0 \int_{0^-}^{t} u(t)\,d\tau$$

$$[y'(t) - y'(0)] + a_1 [y(t) - y(0^-)] + a_2 \int_{0^-}^{t} y(\tau)\,d\tau =$$

$$= b_0 (u'(t) - u(0^-)) + b_1 (u(t) - u(0^-)) + b_0 \int_{0^-}^{t} u(t)\,d\tau \quad \Big/ \int_{0^-}^{t}$$

$$\rightarrow \int_{0^-}^{t} y'(\tau)\,d\tau - \int_{0^-}^{t} y'(0^-)\,d\tau + a_1 \left[ \int_{0^-}^{t} y(\tau) - \int_{0^-}^{t} y(0^-)\,d\tau \right] + a_2 \int_{0^-}^{t} \int_{0^-}^{\tau} y(\lambda)\,d\lambda\,d\tau$$

$$= b_0 \left[ \int_{0^-}^{t} u(\tau)\,d\tau - \int_{0^-}^{t} u(0^-)\,d\tau \right] + b_1 \left[ \int_{0^-}^{t} u(\tau)\,d\tau - \int_{0^-}^{t} u(0^-)\,d\tau \right]$$

$$+ b_0 \int_{0^-}^{t} \int_{0^-}^{\tau} u(\lambda)\,d\lambda\,d\tau$$

$$\Rightarrow y(t) - y(0^-) - y'(0^-)(t - 0^-) + a_1 \left[ \int_{0^-}^{t} y(\tau)\,d\tau - y(0^-)(t - 0^-) \right] + a_2 \int_{0}^{t} \int_{0^-}^{\tau} y(\lambda)\,d\lambda\,d\tau$$

$$= b_0 \left[ u(t) - \cancel{u(0^-)} - \cancel{u(0^-)(t - 0^-)} \right] + b_1 \left[ \int_{0^-}^{t} u(\tau)\,d\tau - \cancel{u(0^-)(t - 0^-)} \right]$$

$$+ b_0 \int_{0^-}^{t} \int_{0^-}^{\tau} u(\lambda)\,d\lambda\,d\tau$$

$t = 0^+$ $\rightarrow y(0^+) - y(0^-) - \cancel{y'(0^-)(0^+ - 0^-)} + \cancel{a_1 \int_{0^-}^{0^+} y(\tau)\,d\tau} - \cancel{a_1 y(0^-)(0^+ - 0^-)}$

U ulazu nema impulsa $\rightarrow$ na izlazu nema skoka

$$+ \cancel{a_2 \int_{0^-}^{0^+} \int_{0^-}^{\tau} u(\lambda)\,d\lambda\,d\tau}$$

rez. nakon $\int$ nema naglu promjenu vrij. $\rightarrow 0^- = 0^+$

$$= b_0 u(0^+) + \cancel{b_1 \int_{0^-}^{0^+} u(\tau)\,d\tau} + \cancel{b_0 \int_{0^-}^{0^+} \int_{0^-}^{\tau} u(\lambda)\,d\lambda\,d\tau}$$

$$\rightarrow \boxed{y(0^+) - y(0^-) = b_0 u(0^+)}$$

još jedan poč. uvjet moramo prevačunati $\longrightarrow y'(0^+)$

$\longrightarrow$ vraćamo se na izraz prije druge integracije

$$y'(t)-y'(0^-)+a_1[y(t)-y(0^-)]+a_2\int_{0^-}^{t} y(\tau)d\tau = b_0u'(t)+b_1u(t)+b_2\int_{0^-}^{t}u(\tau)d\tau$$

$t=0^+$

$$y'(0^+)-y'(0^-)+a_1[y(0^+)-y(0^-)]+\cancel{a_2\int_{0^-}^{0^+} y(\tau)d\tau}$$

nema skokovite promjene

$$=b_0u'(0^+)+b_1u(0^+)+\cancel{b_2\int_{0^-}^{0^+}u(\tau)d\tau}$$

$$\boxed{y'(0^+)-y'(0^-)+a_1[y(0^+)-y(0^-)]=b_0u'(0^+)+b_1u(0^+)}$$

▸ostalo su čisti brojevi koje uvrštavamo

• ako se radi o jednadžbi 3. ili višeg reda:

1) $y(0^+)-y(0^-)=b_0u(0^+)$

2) $y'(0^+)-y'(0^-)+a_1y(0^+)-a_1y(0^-)=b_0u'(0^+)+b_1u(0^+)$

3) $y''(0^+)-y''(0^-)+a_1y'(0^+)-a_1y'(0^-)+a_2y(0^+)-a_2y(0^-)=b_0u''(0^+)+b_1u'(0^+)+b_2u(0^+)$

$\vdots$

$$y^{(N-1)}(0^+)-y^{(N-1)}(0^-)+a_1[y^{(N-2)}(0^+)-y^{(N-2)}(0^-)]+\dots+a_{N-2}[y'(0^+)-y'(0^-)]$$
$$+a_{N-1}y(0^+)-a_{N-1}y(0^-)=$$
$$=b_0u^{N-1}(0^+)+b_1u^{(N-2)}(0^+)+\dots+b_{N-2}u'(0^+)+b_{N-1}u(0^+)$$

$N=2$

Primjer: $y''(t)+2y'(t)+5y(t)=u(t)$ $\quad y(0^-)=-1 \quad y'(0^-)=-1$

Šalabahter: $y(0^+)-y(0^-)=b_0u(0^+)$ $\quad u(t)=5\mu(t)$

$b_0=0$

$y(0^+)-(-1)=0\cdot 5$ $\quad a_1=2$

$b_1=0\rightarrow u'(t)$

$y(0^+)=-1$ $\quad b_2=1\rightarrow u(t)$

! PAZI

2. formula: $y'(0^+)-y'(0^-)+a_1[y(0^+)-y(0^-)]=b_0[u'(0^+)-u'(0^-)]+b_1[u(0^+)-u(0^-)]$

$y'(0^+)-(-1)+2\underbrace{\cancel{[-1-(-1)]}}_{0}=b_0\cdot u(0^+)+b_1u(0^+)$ (jer je step*)

$y'(0^+)+1=\cancel{0\cdot u(0^+)}+\cancel{0.5}\longrightarrow y'(0^+)=-1$

Primjer: naći poč. uvjet za jednadžbu:

$$y'(t) + 2y(t) = -u'(t) + 8u(t) \quad ; \quad u(t) = \mu(t) \quad , \quad y(0^-) = 2 \quad , \quad y(0^+) = ?$$

$a_1 = 2$ $\quad b_0 = -1$

$b_1 = 8$

$$y(0^+) - y(0^-) = b_0 \cdot u(0^+)$$

$$y(0^+) = -1 \cdot 1 + 2 \rightarrow \boxed{y(0^+) = 1}$$

→ budući da je jednadžba prvog reda → samo 1 poč uvjet

Primjer:

$$y''(t) + 2y'(t) + 5y(t) = u(t)$$

$$u(t) = 5\mu(t)$$

$$\left.\begin{array}{l} y(0^-) = -1 \\ y'(0^-) = -1 \end{array}\right\} \quad \begin{array}{l} y(0^+) = -1 \\ y'(0^+) = -1 \end{array}$$

$$y_h(t) = C_1 \cdot e^{(-1+2j)t} + C_2 \cdot e^{(-1-2j)t}$$

$$y_p(t) = 1 \quad , \quad t \geq 0$$

$$\Rightarrow y(t) = C_1 \cdot e^{(-1+2j)t} + C_2 \cdot e^{(-1-2j)t} + 1$$

$$y'(t) = C_1 \cdot (-1+2j) e^{(-1+2j)t} + C_2 (-1-2j) e^{(-1-2j)t}$$

$t = 0^+$

$$y(0^+) = C_1 + C_2 + 1 = -1$$

$$y'(0^+) = C_1(-1+2j) + C_2(-1-2j) = -1$$

$$\Rightarrow C_1 = -1 + j0.75$$

$$C_2 = -1 - j0.75$$

$$\Rightarrow \underbrace{y(t)}_{\text{totalni odziv}} = \underbrace{(-1+j0.75)e^{(-1+2j)t} + (-1-j0.75)e^{(-1-2j)t}}_{\text{prirodni odziv}} + \underbrace{1}_{\text{prisilni odziv}} \quad , \quad t \geq 0$$

→ poč. jednadžba ima prirodne brojeve pa će i rj imati samo treba dalje raspisati da se pokrate dobijemo sinuse i kosinuse

$$y_p(t) = -1\left(e^{(-1+2j)t} + e^{(-1-2j)t}\right) + j \cdot 0{,}75\left(e^{(-1+2j)t} - e^{(-1-2j)t}\right)$$

$$= -e^{-t}\left(e^{2jt} + e^{-2jt}\right) + e^{-t} j0{,}75\left(e^{2jt} - e^{-2jt}\right)$$

$$= -e^{-t}(2\cos(2t)) - 0.75 \cdot e^{-t} \cdot \sin(2t) = -2e^{-t}\cos(2t) - 1.5e^{-t}\sin(2t)$$

$$\Rightarrow \underbrace{y(t)}_{\text{totalni}} = \underbrace{\left[-2e^{-t}\cos(2t) - 1.5e^{-t}\sin(2t)\right.}_{\text{prirodni odziv}} \left. + 1\right] \mu(t)$$

prisilni odziv

mijenja amplitudu → prigušuje

→ polako kroz vrijeme će nestati prirodni i ostaje samo prisilni odziv $(1 \cdot \mu(t))$

## još možemo tražiti odziv NEPOBUĐENOG

NEPOBUĐENI: $y_0(t) = C_1 \cdot e^{(-1+2j)t} + C_2 \cdot e^{(-1-2j)t}$

$y(0^-) = -1 \qquad y(0^+) = -1 \qquad y_0'(t) = C_1 \cdot (-1+2j) e^{(-1+2j)t} + C_2(-1-2j) e^{(-1-2j)t}$

$y_0(0^-) = C_1 + C_2 = -1$

$y_0'(0^-) = C_1(-1+2j) + C_2(-1-2j) = -1$

* radimo samo za homogenim dj jer $u(t) = \emptyset$

$\ldots \Rightarrow \quad C_1 = -0.5 + j0{,}5 \qquad C_2 = -0.5 - j0{,}5$

$y_0(t) = (-0{,}5 + j0{,}5) e^{(-1+2j)t} + (-0.5 - j0{,}5) e^{(-1-2j)t}$

$\Rightarrow \ldots$ opet svedimo do cos i sin $\Rightarrow \boxed{y_0(t) = -e^{-t} \cdot \cos(2t) - e^{-t} \cdot \sin(2t)}$

rješenje za nepobuđeni odziv

## MIRNI ODZIV → poč. uvjeti = 0

$* \; y''(t) + 2y'(t) + 5y(t) = u(t)$

$a_0 = 1 \qquad a_1 = 2 \qquad a_2 = 5 \qquad b_0 = 0 \qquad b_1 = 0 \qquad b_2 = 1$

$y_m(t) = C_1 e^{(-1+2j)t} + C_2 e^{(-1-2j)t} + 1$

$y(0^-) = 0 \qquad y'(0^-) = 0$

$y_m'(t) = C_1(-1+2j) e^{(-1+2j)t} + C_2(-1-2j) e^{(-1-2j)t}$

$y(0^+) - \underbrace{y(0^-)}_{\emptyset} = b_0 \underbrace{u(0^+)}_{5}$ → $y(0^+) = 0$ ostao $\emptyset$

$y'(0^+) - \cancel{\underbrace{y'(0^-)}_{\emptyset}} + \cancel{a_1 [\underbrace{y(0^+)}_{\emptyset} - \underbrace{y(0^-)}_{\emptyset}]} = \cancel{b_0 (\underbrace{u'(0^+)}_{0} - u(0^-))} + \cancel{b_1 [\underbrace{u'(0^+)}_{0} - \underbrace{u(0^-)}_{5}]}$

$y'(0^+) = 0$

$\Rightarrow y_m(t) = C_1 + C_2 + 1 = 0$

$y_m'(0^+) = C_1(-1+2j) e^0 + C_2(-1-2j) e^0 = 0$

$C_1 = -0.5 + j0.25$

$C_2 = -0.5 - j0.25$

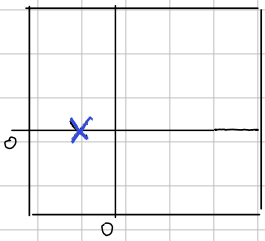
$$\Rightarrow \boxed{\begin{aligned} y_m(t) &= (-0.5 + j0.25) e^{(-1+2j)t} + (-0.5 - 0.25j) e^{(-1-2j)t} + 1 \\ &= \ldots = -e^{-t} \cos(2t) - 0.5 e^{-t} \cdot \sin(2t) + 1, \quad t \geq 0 \end{aligned}}$$

$y(t) = y_0(t) + y_m(t) = [-2e^{-t} \cos(2t) - 1.5 e^{-t} \cdot \sin(2t) + 1] \mu(t)$

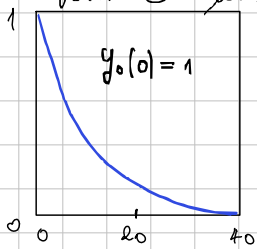
dobili smo isti stvar!

# Odziv neperturbiranog vremenski KONT sustava i vlastite frekv

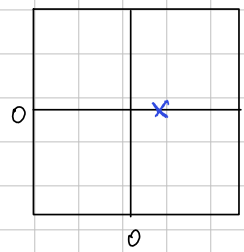
$s = -0.1$



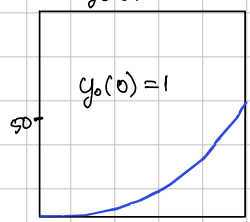

$y_0(t) = e^{-0.1t}\mu(t)$



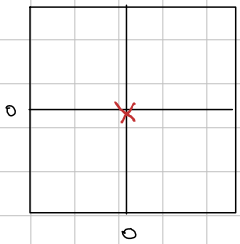

$s_1 = 0.1$



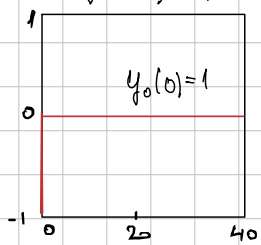

$y_0(t) = e^{0.1t}\mu(t)$



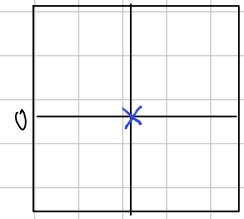

$s_1 = 0$



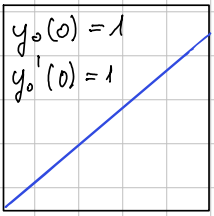

$y_0(t) = \mu(t)$



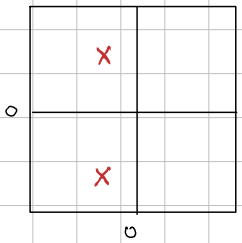

$s_{1,2} = 0$



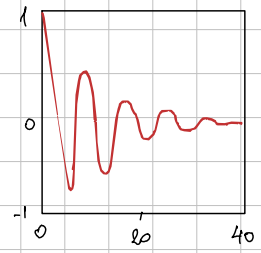

$y_0(t) = (1+t)\mu(t)$



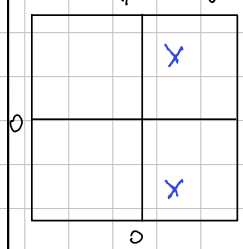

$s_{1,2} = -0.1 \pm j$



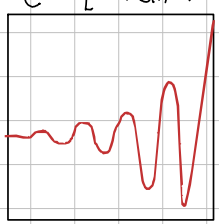

$e^{-0.1t}[0.1\sin(t) + \cos(t)]\mu(t)$



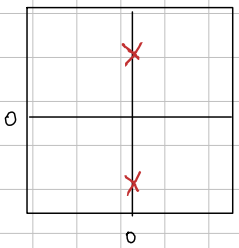

$s_{1,2} = 0.1 \pm j$



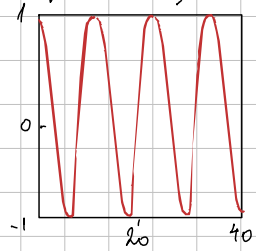

$e^{0.1t}[-0.1\sin(t) + \cos(t)]\mu(t)$



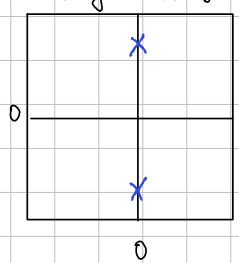
$s_{1,2} = \pm j$

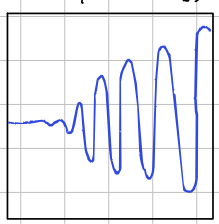



$y_0(t) = \cos(t)\mu(t)$




$s_{1,2} = j$, $s_{3,4} = -j$




$[\cos(t) + 0{,}5t \cdot \sin(t)]\mu(t)$